

Lohit Ashwa Vaswani

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Education

- **University Of Surrey** **Feb 2025 - Present**
Master of Science - Data Science **Guildford, United Kingdom**
Courses: Machine Learning & Data Mining, Data Science Principles and Practices, Natural Language Processing, Cloud Computing.
- **National University of Emerging Sciences – FAST** **Aug 2020 - June 2024**
Bachelor of Science - Artificial Intelligence; CGPA: 3.33 **Karachi, Pakistan**
Courses: Data Structures, Analysis Of Algorithms, Generative Artificial Intelligence, Computer Vision, Machine Learning, Natural Language Processing, Databases, Object Oriented Programming.

Skills Summary

- **Languages:** Python , SQL, JavaScript, C/C++.
- **Frameworks:** Scikit, Pytorch, Open CV, TensorFlow, Keras, Langchain, Llama-index.
- **Structured/Unstructured & Vector DataBases:** Oracle, MySQL, MongoDB, Faiss, Pinecone, ChromaDB.
- **Tools:** Docker, GIT, Tableau, PowerBI, AWS (SageMaker, Lambda, Load Balancing).
- **AI Domains:** NLP, Computer Vision, Generative AI, Recommendation Systems, MLOps
- **Soft Skills:** Communication, Critical thinking, Team collaboration, Continuous learning, Time Management.

Experience

- **Retrocausal — Research Engineer** **Feb 2024 – Jan 2025**
Promoted from Intern to Full-Time Engineer — Remote **Redmond, United States**
 - **Ergonomics Report Generator:** Built an automated Excel reporting tool for ergonomic assessments (REBA, RULA, NIOSH, SNOOK, Hand Strain) integrated with a Django backend via POST APIs. Later modularized as a FastAPI microservice, fully containerized with Docker.
Impact: Cut report generation time by 44%, improving ergonomics workflow and client UX.
 - **Model Optimization (OpenVINO):** Converted YOLOv5 models to INT8-quantized OpenVINO format for edge deployment, significantly boosting inference speed with negligible accuracy loss.
Impact: Achieved 2× faster inference while reducing compute costs by 36%.
 - **Video Captioning System:** Fine-tuned Timesformer+GPT-2 and ViT+GPT-2 models to auto-caption assembly line videos, identifying missing or misinstalled components.
Impact: Reduced assembly errors by 25%, lowering rework costs by 30%.
- **TPS Worldwide — Product Development & ML Engineer** **Jul 2023 – Aug 2023**
Internship — On-Site **Karachi, Pakistan**
 - Developed advanced SQL queries for Oracle databases to support data analysis workflows.
 - **ATM Recommendation System:** Built a ML-powered ATM withdrawal recommender using SVM, XGBoost, and neural networks. Personalized suggestions based on user history, time, and location data.
Impact: Deployed system received 250+ positive reviews from end users.
- **National University of Emerging Sciences (FAST) — Teaching Experience**
Teaching & Academic Support — On-Site
 - **Student Lab Assistant – OOP (Aug 2022 – Dec 2022):** Mentored students on core OOP concepts, helped debug code, and reinforced software development best practices.
 - **Teaching Assistant – AI Programming (Aug 2023 – Dec 2023):** Conducted sessions on AI topics like search algorithms, game theory, and KR&R; assisted students in developing AI models.
 - **Teaching Assistant – Recommendation Systems (Jan 2024 – May 2024):** Guided students in building collaborative filtering and neural recommender models; facilitated discussions on model tuning and ethical implications.

Projects

- **Automating the Evaluation of JEE Report Quality Using LLMs (PoC – UKHSA)** **June 2025 - Sept 2025**
Python, Mistral OCR, Hugging Face Transformers, PyTorch, BERTScore, QWK, JSON Schema
Motivation: Manual JEE evaluations often lack consistency and transparency; this project aimed to create an automated, rubric-based evaluation framework using LLMs to ensure objective, reproducible, and evidence-grounded assessments.
 - Annotated JEE data using a **standardized scoring rubric** with detailed justifications in JSON format for two domains—**Evidence Quality** and **Actionability Quality**—across all technical areas.
 - Used **Mistral OCR** to extract and structure JEE PDFs into a predefined JSON schema for reproducible, machine-readable evaluation.

- Benchmarked **Gemini 2.5 Flash**, **Qwen-3 4B Instruct**, and **Magistral-Medium-Latest** using **QWK** (ordinal agreement) and **BERTScore-F1** (semantic accuracy) against annotations.
- Delivered as a **Proof of Concept (PoC)** for the **UK Health Security Agency (UKHSA)**, demonstrating the feasibility of **LLM-based automated evaluation** for WHO reporting frameworks.

• **E-Commerce Microservices System**

May 2025

FastAPI, MongoDB, Docker, JWT, bcrypt, Sentence Transformers, React.js

Designed and deployed a scalable microservices-based e-commerce platform supporting user accounts, product management, transactions, and recommendations.

- Implemented JWT authentication with bcrypt hashing and role-based access control for security.
- Integrated **all-MiniLM-L6-v2** for semantic product search and personalized recommendations.
- Orchestrated containerized services using Docker Compose for platform-agnostic deployment.
- Conducted load testing with Locust and validated APIs using Postman.
- Built a responsive React.js frontend to deliver a smooth user experience.

• **Revolutionizing Trading Insights with Llama-2**

Aug 2023 – Mar 2024

PyTorch, RAG, Pinecone, Llama-2, LlamaIndex

Created an LLM-powered system to convert complex trading graphs into natural language summaries, improving usability for traders at **Trade for Sight Company**

- Built an ETL pipeline to transform SQL-based numerical data into structured language representations for Llama-2.
- Integrated Pinecone with LlamaIndex for semantic retrieval and hallucination-resistant, grounded output generation.
- Applied Chain-of-Thought prompting to enable logical breakdown of graph data before summarization.
- Resulted in 30% user engagement boost, 50% faster interpretation time, and 70% adoption among active traders.

• **LLAMA-2 Augmented Mental Health Support System**

Dec 2023

PyTorch, Gradio, QLoRA, PeFT, Kaggle GPU P100

Built a virtual therapist using Llama-2 (7B) to deliver real-time mental health support, fine-tuned on the `Amod.mental_health_counseling_conversations` dataset.

- Quantized Llama-2 from 32-bit to 4-bit using QLoRA to reduce memory and computation load.
- Applied Parameter-Efficient Fine-Tuning (PeFT) to train a small subset of weights efficiently.
- Developed an interactive Gradio interface for accessible user interaction.
- Deployed model to Hugging Face: `Lohit20/DepressedLlama-2-7b`.

• **E-commerce Visual Search System Using CLIP + ViT**

Oct 2023 – Nov 2023

PyTorch, Colab GPU T4, Pinecone, Streamlit

Built a multi-modal visual search system to help users find products using images or text, enhancing e-commerce navigation without relying on exact keywords.

- Combined CLIP and ViT for robust image-text embedding and semantic similarity matching.
- Used Pinecone to store and retrieve vector embeddings via cosine similarity for relevant product recommendations.
- Enabled dual search modes—image or text input—for flexible product discovery.
- Developed a user-friendly Streamlit interface to facilitate real-time interaction.

• **Movie Recommendation System (Multi-Armed Bandits + Content-Based Filtering)**

Mar 2023

Python, Scikit-learn

Built a hybrid movie recommendation system combining reinforcement learning and content-based filtering to deliver personalized suggestions and address the cold start problem.

- Implemented collaborative multi-armed bandits for adaptive exploration of evolving user preferences.
- Applied content-based filtering using item metadata to recommend similar movies based on user history.

• **Stock Price Prediction Using LSTM**

Dec 2022

Python, Selenium, LSTM, Tableau

Built an LSTM-based model to forecast stock prices using historical data, achieving a Mean Squared Error 0.0680.

- Scraped real-time stock data from NASDAQ using Selenium for time-series modeling.
- Trained LSTM network to learn temporal dependencies in stock movements, improving prediction accuracy.
- Developed a Tableau dashboard to visualize trends, identify patterns, and display future projections.

Relevant Certification

- Orchestrating Workflows for GenAI Applications (DeepLearning.AI)- July 2025.
- Generative AI Concepts + Introduction to LLMs in Python (Data Camp)-Mar 2024.
- TensorFlow for Artificial Intelligence, Machine Learning & Deep Learning (DeepLearning.AI)-Aug 2023.
- Ensemble Methods in Python(Data Camp)-Nov 2022.

Honors and Awards

- Thrice Honored in Deans list Fast University (Spring 2023, Spring 2021, Spring 2024).
- Data Science PROCOM Competition-7th/30+teams (Spring 2023).